

RENSTON JAKE FERNANDES, Ph.D

Postdoctoral Researcher | Energy Engineering | Process Modelling & Simulation |
Thermal Biomass Conversion | Techno-Economic Analysis | Life Cycle Assessment

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RESEARCH PROFILE

Interdisciplinary process-systems, energy engineering, and sustainability researcher with a **Ph.D. in Environmental Science and Engineering** and a strong publication record in the environmental assessment of emerging technologies. Integrates **Aspen Plus process simulation, techno-economic analysis (TEA), ISO-aligned life-cycle assessment (LCA), experimental research, and uncertainty analysis** to support process optimization, project development, and industrial scale-up. Research experience spans carbon capture and utilization, waste management and resource recovery, circular economy, biomass gasification, pyrolysis, mineral carbonation, industrial decarbonization, hydrogen, ammonia, and waste-heat recovery.

Core Competencies: Environmental assessment • Prospective and ex-ante LCA • Process engineering and simulation • Mass and energy balances • Techno-economic assessment • Process optimisation • Waste management and resource recovery • Carbon capture and utilisation • Project development and project management • Scenario and uncertainty analysis • Data analysis skills • Cross-functional collaboration

Current Toolkit: Environmental LCA • Foreground LCI development • Techno-economic analysis • Scenario analysis • Sensitivity and Monte Carlo uncertainty • Aspen Plus • GREET • GaBi (Sphera) • Python pipelines for reproducible sustainability assessment

EDUCATION

King Abdullah University of Science and Technology, Doctor of Philosophy (Ph.D) | Environmental Science and Engineering 11/2021 – 05/2026 | Kingdom of Saudi Arabia

Dissertation Topic: "**Circular Carbon E-Fuels from Biomass: Techno-Economic, Environmental, and Combustion Assessment Towards a Nation-Centric Sustainable Fuel Strategy.**"

University of Sussex 09/2018 – 01/2020 | Brighton, United Kingdom
Master of Science (M.Sc) | Engineering Business Management

Master's Dissertation: "**Energy from Wastewater – An Economic Assessment of Wastewater in Skyscrapers.**"

Degree: Distinction (**top 1% of class; Sussex India Scholarship recipient**)

NMAM Institute of Technology 08/2014 – 06/2016 | Mangalore, India
Master of Technology (M.Tech) | Energy Systems Engineering

Master's Dissertation: "**Experimental Investigation of Cycle-by-Cycle (CBC) Fluctuations in a 4-Stroke SI Engine using Gasoline-Ethanol Blends.**"

Degree: Distinction (**top 1% of class**)

Sahyadri College of Engineering and Management 09/2010 – 06/2014 | Mangalore, India
B.E (Bachelor of Engineering) | Mechanical Engineering

Bachelor's Research Project: "**Mechanically Operated Wheelchair cum Stretcher**"

Degree: First Class

PROFESSIONAL RESEARCH EXPERIENCE | WORK HISTORY

King Abdullah University of Science and Technology (KAUST) 06/2026 – Present | Kingdom of Saudi Arabia
Researcher

Research conducted under **Prof. Bassam Dally** on industrial decarbonisation, thermochemical conversion, and mineral carbonation. (**The Renewable and Sustainable Energy Technology Group**). Contributing to project development, project management, experimental planning, process-data analysis, progress coordination, and delivery across multidisciplinary research activities.

Oxyfuel cement production and CO₂-to-fuels integration

- Developing and validating an **industrial-scale Aspen Plus model of an oxyfuel cement plant**, encompassing staged preheating, calcination, clinker formation, gas recycle, and CO₂-rich exhaust handling.
- Integrating **captured kiln CO₂ with e-fuel and sustainable aviation fuel pathways**; translating process outputs into conversion, energy-intensity, utility-demand, and carbon-management indicators.

Oil-shale pyrolysis modelling and techno-economic assessment

- Reconstructed and validated a three-stage Aspen Plus pyrolysis flowsheet with non-conventional solids; yield, stoichiometric, and equilibrium reactors; gas-solid separation; recycle loops; and energy integration.
- Building a transparent TEA and cash-flow framework to quantify CAPEX, OPEX, product value, energy demand, coproduct-credit scenarios, and economic sensitivity.

Brine and cementitious mineral-waste carbonation

- Executing and analyzing desalination-brine carbonation experiments with cement kiln dust and red mud; evaluating pH, TDS, gas-phase CO₂ response, repeatability, and solid-liquid evidence for carbon uptake.
- Undertaking hands-on training in **X-ray diffraction (XRD)** and **X-ray fluorescence (XRF)** to support phase and elemental characterization of carbonated solids.

King Abdullah University of Science & Technology

11/2021 – 05/2026 | Kingdom of Saudi Arabia

PhD Researcher

Affiliated with the **Clean Energy Research Platform (CERP)**, formerly **Clean Combustion Research Center (CCRC)**, under **Prof. James W. G. Turner** at the **Advanced Vehicle Propulsion Laboratory (AVPL)**.

PhD Thesis Objective: "**Circular Carbon E-Fuels from Biomass: Techno-Economic, Environmental Sustainability, and Combustion Assessment Toward a Saudi-Centric Sustainable Fuel Strategy**"

- Conducted comprehensive **well-to-tank LCA** and **TEA** of **synthetic e-fuels** derived from regionally dominant agricultural biomass for conventional and novel engine applications.
- Developed integrated **Aspen Plus models** for biomass-to-ethanol fermentation and e-methanol production pathways, extracting performance indicators including conversion efficiency, energy intensity, CO₂ capture rate, and cost drivers across multiple regional feedstocks.
- Conducted well-to-tank life cycle assessment using **GREET**, **GaBi (Sphera)**, **GLAM**, and **ReCiPe** methodologies, contributing to peer-reviewed publications.
- Performed multi-scenario **techno-economic modelling** of integrated biofuel and e-fuel plants to evaluate commercial feasibility, cost drivers, and sustainability trade-offs.
- Built reproducible **Python workflows** for **scenario analysis**, sensitivity analysis, KPI generation, and **Monte Carlo uncertainty propagation**.
- Experimentally tested **hydrous ethanol-methanol blends** in a **CFR engine** and applied structured experimental design to assess the effects of fuel blend composition and operating conditions on combustion and emissions monitoring using **FTIR-based profiling**.

Collaboration 1: Hydrogen-Fueled Variable Compression Ratio (VCR) HCCI Engine.

- Investigated and published **two co-authored peer-reviewed studies** on **hydrogen as a zero-carbon fuel** in a modified single-cylinder CFR engine under HCCI mode across compression ratios (14:1 to 18:1) and intake temperatures (110–150°C).
- **Study 1:** Investigated the effect of variable compression ratio (14:1–18:1) on hydrogen HCCI combustion in a modified CFR engine. Analyzed combustion phasing (CA₅₀), net IMEP, thermal efficiency (~33%), and emissions including near-zero NO_x under ultra-lean $\lambda = 4.8$ conditions.
- **Study 2:** Studied the influence of intake temperature (90–150°C) on hydrogen HCCI performance at fixed CR (17:1). Demonstrated expansion of lean limit to $\lambda = 5$, observed trade-offs in heat release, combustion duration, and knock sensitivity. Achieved thermal efficiency up to 37.9% and defined optimal operating conditions for future hydrogen engine design.

Collaboration 2: Life cycle assessment of ammonia and hydrogen as alternative fuels for marine internal combustion engines.

- Conducted a comprehensive life cycle assessment (LCA), from a **well-to-wheel perspective** for engine-integrated reformer systems.
- The fuel cycle was split into two phases including production of fuel or well-to-tank (WTT), blue ammonia via SMR and CCS, green hydrogen via electrolysis, feedstock transportation, and onboard storage.

Collaboration 3: Waste Heat Recovery from Marine and Truck Engines – A Joule Cycle-Based Approach.

- Conducted and published a comprehensive **Techno-Economic Analysis** comparing Open Joule Cycle and Organic Rankine Cycle (ORC) systems for energy recovery in heavy-duty vehicles.

Srinivas Institute of Technology

07/2016 – 05/2018 | Mangalore, India

Teaching Faculty - Department of Marine and Mechanical Engineering

Interdisciplinary Teaching & Curriculum Development

- Lecturing and tutoring modules including **Fluid Mechanics** and **Non-Conventional Energy Sources**, integrating practical and theoretical knowledge.
- Developed curricula incorporating modern engineering tools such as **Computer-Aided Machine & Marine Drawing**, ensuring alignment with real-world engineering applications.

Research Mentorship in Renewable Energy

- Demonstrated the ability to supervise undergraduate research projects in **waste-plastic-to-fuel conversion**, **alternative fuel systems**, and **engine combustion**, guiding students through project planning, experimental work, data analysis, technical reporting, and final presentation.
- **Led two state-funded (KSCST) research projects:** Waste plastic-to-fuel conversion, contributing to sustainability and circular economy efforts. Heavy Fuel Oil (HFO) blended with cooking oil, providing insights into alternative fuel viability.

Administrative & Regulatory Expertise

- Managed the **procurement of project approvals** and clearances from governmental agencies, ensuring compliance with regulatory standards.
- Conducted detailed analyses of departmental documents for accurate distribution and archival, improving institutional efficiency.
- Facilitated **student-industry interactions for research funding** and collaboration, emphasizing the practical applicability of academic research.

SKILLS

Process Engineering and Chemical Process Simulation – Expert

- **Aspen Plus** (process flow modeling, reactors, separation systems, recycle loops).
- **Aspen Custom Modeler**
- **Aspen HYSYS**
- **Engineering Equation Solver (EES)** – for cycle analysis and thermodynamic calculations.

Techno-Economic Analysis and Market Feasibility Analysis of Emerging Energy Systems – Expert

- **Capital cost estimation** (PEC, CAPEX, OPEX using CEPCL indices)
- **Discounted cash flow models** (NPV, IRR, Payback Period, LCOM)
- **Monte Carlo** simulations for uncertainty and sensitivity analysis (Coding with **Python**) and **Excel spreadsheets**.
- **Scenario building** and benchmarking with literature.
- Skilled in **Python** for techno-economic data analysis and risk assessment.

Project Development, Project Management and Scientific Communication – Expert

- Excellent **written and effective communication skills**, demonstrated through **technical reports**, **peer-reviewed manuscripts**, **conference abstracts**, and **publication-ready academic material**.
- Excellent **presentation skills**, with **experience presenting data** and research findings to academic, industrial, and **wider professional audiences**
- Strong **literature review**, **research design**, **data collection**, and **data interpretation skills**.
- Creative approach to **problem-solving** and development of new concepts.

Teaching and Mentoring – Proficient

- **Mentoring** of students and provide guidance to interns by organizing training activities.
- **Collaborative** work in multidisciplinary teams.
- Workshop and **training support**.
- Project planning and **proposal writing support**.

Life Cycle Assessment – Expert

- LCA software: **GREET**, **GaBi (Sphera)** & **Simapro**
- Methodology: **ReCiPe 2016**, **GLAM**
- Predictive modeling using **NSGA-II** for environmental impact analysis, process optimization and optimization strategies.

Data Analysis and Programming Skills – Expert

- Python (data acquisition and processing, Monte Carlo, pandas, NumPy, matplotlib, statistical analysis)
- Excel (advanced formulas, TEA model building).
- MATLAB for post-processing of experimental data.

Interdisciplinary, Soft and Interpersonal Skills

– Expert

- **Strong organized collaboration skills, analytical ability**, and critical thinking skills, with a **structured, evidence-based, and solution-oriented approach** to complex research problems.
- **Ability to work independently**, as well as **cross-functionally** in an interdisciplinary, fast-paced, and dynamic team environment.
- **Resilient, adaptable, and self-directed researcher**, committed to maintaining confidentiality and **delivering effectively** under evolving research demands.
- **Cooperative and collaborative team player** with emerging **leadership skills**, problem-solving skills, and research delivery.
- **Highly motivated professional** with strong self-management, initiative, and a proactive approach to problem-solving.
- **Demonstrated reliability** in research delivery through research outputs and peer-reviewer for several high-impact journals.

Materials Characterization and Analytical Techniques – Expert

- **X-ray Diffraction (XRD)** – hands-on training in phase identification and characterization of carbonated mineral solids.
- **X-ray Fluorescence (XRF)** – hands-on training in elemental/compositional characterization of cementitious and mineral materials.
- **Fluorescence Spectroscopy – Agilent Cary Eclipse Fluorescence Spectrophotometer** – sample analysis and fluorescence measurement.

★ INTERNSHIPS AND SUMMER PROGRAMS

Aramco (Saudi) - Summer Research Intern

06/2025 – 08/2025 | Ras Tanura, Saudi Arabia

Techno-Economic Analysis of a Novel Dual PSA-Based Flue Gas Separation System for E-Methanol Production

- Conducted a **techno-economic evaluation** of a **patented dual-PSA flue gas separation and purification** concept integrated with CO₂ hydrogenation to methanol using **Aspen Plus**.
- Developed a **scenario matrix** varying energy demand, conversion efficiency, process parameters and market drivers, and quantified uncertainty from hydrogen price, electricity cost, and catalyst life.
- Evaluated **commercial feasibility**, scenario-dependent **business opportunities**, and **scale-up implications** for **industrial deployment**.
- Prepared **decision-support summaries** for technical stakeholders, translating model outputs into concise evidence for applied industrial and non-academic use.

Radboud University Summer School - Summer Research Student

07/2024 – 07/2024 | Nijmegen, Netherlands

Advances in Environmental Life Cycle Modeling

- Gained advanced training in **life cycle assessment**, including **inventory modelling**, **impact assessment**, **uncertainty analysis**, and **spatially differentiated environmental modelling**.
- Worked with modelling approaches relevant to **hydrology**, **ecology**, **pollutant impact assessment**, and broader **natural sciences-based environmental evaluation**.
- Studied **attributitional and consequential LCA**, allocation issues, and emerging approaches in circular economy and recycling assessment.
- Developed LCA models using **MATLAB** and **Excel**, applying **LC-IMPACT** and **GLAM** for spatially differentiated impact assessment.
- Built foundational exposure to impact category modelling relevant to **toxicity-oriented** and **pollutant-related** sustainability interpretation.

Lamina Suspension Products Ltd - Graduate Summer Intern

06/2015 – 09/2015 | Mangalore, India

Leaf Spring Energy Performance Analysis

Conducted an LCA-related assessment in a **leaf spring manufacturing** setting, gaining early industrial exposure to **manufacturing systems**, **materials processing**, **heat-treatment operations**, and **process-level energy efficiency monitoring** in a leaf spring production environment.

Process Calibration & Monitoring

- Conducted **heat pumps calibration** and **oil flow monitoring** in heating furnaces, showcasing attention to data accuracy and process monitoring, essential for Life Cycle Assessment (LCA) studies.

Performance Optimization

- Designed and implemented an **employee evaluation process**, leading to measurable **performance improvements** and showcasing skills in **process efficiency and system optimization**.

Zhejiang University - Research Trainee

11/2025 – 11/2025 | Zhejiang, China

9th International Training Workshop on Waste-to-Energy

- The training covered advanced waste management and incineration, **AI-enabled smart WtE**, **Waste-to-X system planning and optimization**, MSW combustion fundamentals, and emissions control technologies (incl. dioxins and pollution control).
- Participated in sessions highlighting **digital-twin-supported plant performance** and operations and broader international perspectives on WtE policy and deployment.

📄 SELECTED RECENT RESEARCH OUTPUTS

Quantifying the potential of Saudi date palm fruit and its waste for e-methanol production using integrated process modelling, techno-economic analysis and life-cycle assessment.

10/04/2026

Elsevier

Journal: Biomass and Bioenergy (Under Review)

Pre-print DOI: <https://dx.doi.org/10.2139/ssrn.6569602> 

Authors(s): **Renston Jake Fernandes**, Mohammad Raghieb Shakeel, Ducduy Nguyen, Hong G. Im, James W. G. Turner

A Joule cycle-based approach to recover waste heat from marine and heavy-duty vehicles

04/10/2025

Elsevier

Journal: Energy

DOI: <https://doi.org/10.1016/j.energy.2025.138755> 

Author(s): Sreenivasa Rao Gubba, Raghu Vamsi Kodaboina, Kesty Y. Kenkoh, **Renston Jake Fernandes**, James W. G. Turner

- Techno-Economic Analysis of an Integrated E-Methanol Production Plant Utilizing First and Second Generation Biomass Resources** 16/06/2025
SAE International
Journal: SAE Technical Paper
DOI: <https://doi.org/10.4271/2025-01-0182> 
Author(s): Renston Jake Fernandes, Mohammad Raghieb Shakeel, Ducduy Nguyen, Hong G. Im, James W.G. Turner
- Life cycle assessment of ammonia and hydrogen as alternative fuels for marine internal combustion engines** 25/03/2025
Elsevier
Journal: International Journal of Hydrogen Energy
DOI: <https://doi.org/10.1016/j.ijhydene.2025.02.331> 
Author(s): Ducduy Nguyen, Renston Jake Fernandes, James WG Turner, David Robert Emberson
- Evaluating Regional Sustainability of E-Methanol Production through GLAM-Based Life Cycle Impact Assessment** 01/01/2025
Elsevier
Journal: Procedia CIRP
DOI: <https://doi.org/10.1016/j.procir.2025.01.057> 
Author(s): R.J. Fernandes, D. Nguyen, M.R. Shakeel, J.W.G. Turner
- Advanced Biomass Conversion: Sustainable e-Methanol Production with Enhanced CO2 Utilization** 09/04/2024
SAE International
Journal: SAE Technical Paper
DOI: <https://doi.org/10.4271/2024-01-2449> 
Author(s): Renston Jake Fernandes, Ducduy Nguyen, Mohammad Raghieb Shakeel, James W.G. Turner
- Variable compression ratio hydrogen-fueled homogeneous charge compression ignition engine** 28/08/2023
SAE International
Journal: SAE Technical Paper
DOI: <https://doi.org/10.4271/2023-24-0067> 
Author(s): Duc Duy Nguyen, Renston Fernandes, James WG Turner
- Investigating the effect of intake temperature on hydrogen HCCI engine combustion and performance characteristics** 04/12/2023
Taylor & Francis
Book: Powertrain Systems for a Sustainable Future
DOI: <http://dx.doi.org/10.1201/9781032687568-23> 
Author(s): Ducduy Nguyen, Renston Fernandes, James W.G. Turner

PEER-REVIEW SERVICE

Verified peer reviewer with **19 completed manuscript reviews** across **eight international journals**:

- **International Journal of Hydrogen Energy** – 7 reviews
- **Fuel Processing Technology** – 3 reviews
- **Fuel** – 3 reviews
- **Bioresource Technology Reports** – 1 review
- **Energies** – 2 reviews
- **Energy Strategy Reviews** – 1 review
- **Energy** – 1 review
- **Journal of CO2 Utilization** – 1 review

CONFERENCES & PRESENTATIONS - 2025

- 12th International Conference on Life Cycle Management** 09/2025 | Palermo, Italy
 Poster Presenter - "Life Cycle Assessment of E-Methanol Production from Dates"
- Powertrains, Energy and Lubricants (PEL) Conference** 06/2025 | Busan, South Korea
SAE International (KSAE)
 Oral Presenter - "Techno-Economic Analysis of Integrated E-Methanol Production Systems Using First- and Second-Generation Biomass"

33rd EUBCE 2025 – European Biomass Conference & Exhibition

06/2025 | Valencia, Spain

Oral Presenter - "Circular Carbon Biofuel and E-Fuel Production from Saudi Arabian Date Palm Biomass: A Process Simulation-Based Study"

13th International Conference "Fuel Science: From Production to Propulsion"

06/2025 | Aachen, Germany

RWTH Aachen University

Oral Presenter - "Sustainable E-Methanol Production using Corn and its Waste in Renewable Energy Systems".

32nd CIRP Conference on Life Cycle Engineering

04/2025 | Manchester, United Kingdom

University of Manchester

Oral Presenter - "Evaluating Regional Sustainability of E-Methanol Production through GLAM-Based Life Cycle Impact Assessment".

CORE PHD LEVEL COURSES

ME 305B - Advanced Computational Fluid Dynamics

08/2022 – 12/2022

King Abdullah University of Science and Technology

ME 377 - Advanced Internal Combustion Engines

01/2022 – 05/2022

King Abdullah University of Science and Technology

ENSE 344 - Modeling Bioprocess and Environmental Engineering

08/2023 – 12/2023

King Abdullah University of Science and Technology

ACCOLADES AND SCHOLARSHIPS

KAUST Ph.D Fellowship, King Abdullah University of Science & Technology (KAUST)

10/2021

Thuwal, Saudi Arabia

Awarded a fully funded Ph.D position at the Clean Energy Research Platform (CERP), King Abdullah University of Science & Technology (KAUST).

Sussex India Scholarship, University of Sussex

08/2018

Brighton, United Kingdom

Competitive merit-based scholarship awarded for outstanding academic performance during M.Sc. at the University of Sussex.

Research Grant (2016-2018), Karnataka State Council for Science & Technology (KSCST)

06/2016

Mangalore, India

Graduate Student Research Grant (2017-2018)

- Secured state funding for the project "Experimental Investigation of Performance Parameters Using HFO Blended with Cooking Oil", contributing to alternative fuel research and sustainability.

Graduate Student Research Grant (2016-2017)

- Awarded state funding for the project "Conversion of Waste Plastic into Fuel Using Heliostat", aligning with circular economy principles and waste-to-energy conversion.

LANGUAGES & COMMUNICATION SKILLS

English Language – Fluent

Hindi – Highly proficient

Konkani – Highly proficient